INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

CHARGERS, BATTERY, LIFT TRUCK AND PALLET TRANSPORTER BATTERY SERVICE

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This specification covers solid-state battery chargers for charging valve regulated lead-acid (VRLA) and flooded lead-acid batteries used for propulsion and operation of industrial type lift trucks and pallet transporters aboard U.S. Navy ships and shore stations.

2. CLASSIFICATION. Battery chargers shall be classified as follows:

2.1 Type.

Type 1 – Single unit for charging 12-cell, 18-cell or more (24 and 36 volts nominal minimum range); 600 to 1200 ampere-hour capacity batteries; flooded, VRLA, absorbent glass mat (AGM) or gelled electrolyte (GEL) chemistries.

Type 2 – Dual unit for charging 12-cell, 18-cell or more (24 and 36 volts nominal minimum range); 600 to 1200 ampere-hour capacity batteries; flooded, VRLA, absorbent glass mat (AGM) or gelled electrolyte (GEL) chemistries.

3. SALIENT CHARACTERISTICS.

3.1 Commercial standard. The charger shall meet UL 1564 except as specified herein.

3.2 Direct current (DC) output characteristics. The output voltage and current of the charger shall be within the minimum ranges required for batteries listed for Types 1 and 2 chargers (see 2.1). Additional capability at higher voltages is acceptable, but not required, if standard on the commercial-off-the-shelf unit.

3.3 Operating and display features. The required minimum operating and display features shall be as follows:

a. Message display with a minimum of 16 characters for display of voltage, current, temperature, and other parameters of interest to the user including battery identification, history, charge profile, and data provided by the charger or monitoring system module as specified herein.

b. Operate with the charger manufacturer's Battery Identification (BID) system (see 3.3.3) to provide information such as battery type identification, serial number, battery parameters, charge profile, and charge and discharge history if available.

c. Automatic recognition of battery nominal voltage (24, 36 volts) and selection of charging voltage prior to start of charging.

d. Automatic temperature recognition and compensation of charging voltage based on actual battery temperature during charging.

e. Automatic alarm and charge start disabling due to too-high battery temperature for normal charging.

f. Manual override of alarm and charge disabling features.

g. When actual battery temperature is not available, manual entry of a starting battery temperature shall be accommodated.

h. When no charging profile is selected, manual entry of output voltage and current limit steps in a constant current (I), constant voltage (E), and constant current (I) (I-E-I) shall be accommodated.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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3.3.1 Additional features. Commercial features such as those listed below, but not required herein, may be provided as part of a standard design. These features shall not interfere with required operational characteristics specified herein.

a. Input power failure recovery allowing return to correct charge state after electrical power has been restored instead of restarting the charge from the beginning in automatic mode.

b. Selectable time delay feature for cool-down of a hot battery.

c. A computer interface connection to a printer or laptop computer.

d. Selectable fast charge feature for charging lead-acid flooded, VRLA, and GEL batteries when allowed by the battery manufacturer.

3.3.2 Alternating current (AC) input characteristics. The charger shall operate from a Type I, 450-volt AC, 50- or 60-hertz, three-phase delta, ungrounded power source in accordance with MIL-STD-1399-300.

3.3.3 BID and monitoring system. The BID system shall be reprogrammable. The system shall work with the battery charger and digital interface devices to provide tabulation of battery charge and discharge history either maintained in the BID device installed on the battery or in the charger for download to a laptop computer or other similar device. The BID system shall have the minimum characteristics as follows:

a. Battery identification (VRLA, GEL, or flooded type), manufacturer, serial number or other identification, and be reprogrammable

b. Battery voltage and temperature provided to the charger.

c. If battery current is a BID data feature, the shunt shall be bolted in line with the positive or negative lead in VRLA and flooded batteries.

d. BID module data interface between the charger and laptop computer shall be universal serial bus (USB) port connection, RS-232 connection, or equal. BID module data transfer to handheld digital device shall be by USB port connection or by Bluetooth® wireless technology, radio power class 3, in accordance with Bluetooth® Specification 2.1 + EDR, or equal wireless technology.

3.4 Interface requirements.

3.4.1 Dimensions. Battery chargers shall not exceed the following dimensions:

Width: 29.5 inches

Height: 38.0 inches

Depth: 26.0 inches

Mass: 400 pounds for a Type 1 charger and 900 pounds for a Type 2 charger

3.4.2 Input phase balance and grounding. Input supply current shall be balanced to the extent that the current in any phase shall not differ by more than 10 percent from the average current in the three phases. The battery charger shall not establish an electrical ground or require grounding, either directly or by means of filter capacitors, of the power source in accordance with the system grounding requirements of MIL-STD-1399-300 for user equipment.

3.4.3 Charger output cabling. Each charger shall be fitted with a cable and connector assembly to connect the battery charger output terminals to the battery connector.

3.4.3.1 Cable. The battery charger output cable shall conform to UL 1581, Type W, two-conductor, AWG 2/0 portable power cable. The cable shall be 14 feet in length and shall contain a minimum of two conductors unless otherwise stated in the contract or order (see 7.2).

3.4.3.2 Connectors. The output connector for the battery chargers to the battery shall be Anderson Power Products SBX 350, or equal, color gray, or Euro style connectors with auxiliary contacts for BID module connection and shall be provided by the vendor when specified (see 7.2). Adapters needed to connect to batteries with connectors of a different color and physical shape, usually associated with dual voltage chargers, shall be supplied when specified (see 7.2).

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3.5 Environmental requirements.

3.5.1 Shock. The battery charger shall be Grade B shock qualified for light or medium weight classification as required by the weight of the charger in accordance with MIL-S-901.

3.5.2 Ambient temperature. The battery chargers shall operate in an ambient temperature range of 32 °F to 120° F.

3.5.3 Electromagnetic interference (EMI). The charger shall meet the requirements of MIL-STD-461, CE101 (submarine), CE102, CS101, CS114, CS116, RE101 and RE102. The acceptable minimum is to meet the requirements of MIL-STD-461, CE101 (submarine), CE102, CS101, CS114, CS116, RE101 through the use of the supplier's input filter. To meet the requirements of RE102, the supplier may need to provide additional shielding behind the display and tactile switch button modules to prevent radiated emissions from passing through the display device. This shielding may take the form of a sheet metal enclosure, foil, screening, spray-on conductive metallic coating, or other means. The use of shielded output cables is permitted.

3.5.4 Corrosion-resistance. All metal parts of the battery chargers shall be fabricated from corrosion-resistant materials, or have an acceptable corrosion-resistant treatment. Circuit cards and other internal components shall be protected from salt-air environment.

3.5.5 Ventilation. Ventilation of the battery chargers shall be natural draft or with fan assist. Cooling shall provide adequate flow such that under normal operating conditions, the maximum temperature rating of any component is not exceeded. Cooling fans shall have a continuous duty cycle. Filters are required for forced ventilation and shall be reusable and washable.

3.5.6 High-temperature protection. Protection shall be provided to avoid component damage from high temperatures that may arise upon failure of ventilation or from operation in a high-temperature environment. Such protection shall automatically de-energize the battery charger at the DC output. The AC input shall remain energized to operate the charger and its cooling fans.

3.6 Operational requirements.

3.6.1 Current adjustment. The DC output current of the battery chargers shall be capable of automatic and manual adjustment to provide different maximum output currents for different ampere-hour capacity batteries.

3.6.2 Operation (battery charging). Battery chargers shall automatically regulate the output current and voltage for the type of battery being charged as recognized by the BID system or as set into the charger manually by the operator. Normal charges shall be completed in 8 to 8.5 hours from 80 percent depth of discharge (DOD) and equalizing charges shall be completed in an additional 3 hours using electronic timers or DV/DT technology to terminate the equalizing charge. The charger shall automatically adjust the charging voltage limit for battery temperature to prevent excessive gassing or undercharging of the battery.

3.6.2.1 Fast charging. When provided, the fast charging feature (see 3.3.1) shall be capable of charging a discharged battery in 4 hours or less. The fast charging feature shall not be automatic and shall be manually selectable in order to maintain guarantees by battery suppliers to the U.S. Navy who prohibit the use of fast charging of VRLA batteries.

3.6.3 Temperature compensation. Charging voltage shall be automatically compensated for variations in actual battery temperature by adjusting the voltage -3 to -4 millivolts per cell for each 1 °F differential in temperature from 80 °F.

3.6.4 Mis-operation. Neither the battery charger nor the operator shall sustain damage/injury when the charger is:

a. Connected with incorrect polarity to any battery.

b. Connected to any battery, regardless of setting of control switches, and regardless of whether the AC input is energized.

c. Disconnected from any battery while charging is in progress.

d. Energized without having the output cable connected to a battery.

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3.6.5 Remote sensing. Remote sensing of battery voltage and battery temperature shall be provided for battery charger operation to compensate for cable voltage drop and provide accurate temperature compensation of charging voltage.

3.6.6 Power switch. The battery chargers shall be provided with a power switch to provide for on-off operation of the charger at the AC input circuit.

3.6.6.1 Power on indication or light. Battery chargers shall be provided with a light or indication to show when the battery charger is energized.

3.6.7 Indicating devices. The battery chargers shall be furnished with indicating devices as an integral part of the equipment. The devices shall provide, as a minimum, an indication of output current and output voltage. These indications may be integrated into the display system required in 3.3.

3.6.8 Overload protection. Overload protection shall be provided in the DC load circuit. The overload protection shall also serve to protect battery charger components in the event of reversed polarity battery connections.

3.6.9 Mounting. Battery chargers shall be deck or bulkhead mounted. The charger enclosure shall be grounded/bonded for personnel safety in accordance with MIL-STD-1310.

3.6.10 Lifting during installation. Battery chargers shall be capable of being lifted while being handled during installation. Straps or attachment points for lifting shall be located such that when lifting vertically the battery charger does not tilt more than 5 degrees from the vertical.

3.7 Prohibited components. The following components shall not be used:

a. Variable transformers (typically “Variacs”).
b. Meter relays.
c. Electron tubes or other thermionic devices.
d. Auto-transformers or open-delta connected transformers.

3.8 Maintainability. Maintenance and repair of battery chargers shall be accomplished within 1 hour.

3.9 Identification plates. The battery chargers shall be provided with a non-corrosive, permanently installed identification plate, providing the following information:

a. Nomenclature.
b. Type 1 or Type 2.
c. National stock number (NSN).
d. Component identification number (CIN).
e. Contract or order number.
f. Electrical characteristics.
g. Manufacturer’s serial number.
h. Manufacturer’s name and address.

3.9.1 Warning plate. The battery charger shall be provided with a "WARNING" plate that alerts personnel of any potential danger of the battery charger and specifies the safety precautions that shall be observed while operating it. The specifics of potential hazards shall be identified along with directions on what to do if a mishap occurs.

3.10 Prohibited materials. Components shall be free of asbestos, mercury and mercury compounds, polyvinyl chloride (PVC), cadmium, and any ozone depleting chemicals.

4. REGULATORY REQUIREMENTS. The use of recovered materials is encouraged to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the battery charger.

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5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producers own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Conformance testing. When specified (see 7.2), the battery charger submitted by the manufacturer as meeting the salient characteristics of this CID shall be tested by the government or a qualified commercial laboratory (see 7.2 for conformance with the shock requirements of 3.5.1 and the EMI requirements of 3.5.3). This testing is required to determine the need for additional measures to reduce emissions or susceptibility.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Bluetooth®. Bluetooth® specifications are available online from http://www.bluetooth.com/Bluetooth/SIG/.

7.1.2 Defense specifications and standards. Defense specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil.

7.1.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at http://www.arnet.gov/far.

7.1.4 Underwriters Laboratories, Inc. (UL). UL standards are available from COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515 or online at www.ul.com.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number.
b. Number and type of units required (see 2.1).
c. Number of BID system modules required (see 3.3.3).
d. Length of charging cable if other than 14 feet in length (see 3.4.3.1).
e. Type of battery cable connector adapter (see 3.4.3.2).
f. Connectors with auxiliary contacts for BID module connection or direct sensor information (see 3.4.3.2).
g. Location of EMI and input power waveform testing (see 5.2).
h. Location of shock testing (see 5.2)
i. Packaging requirements (see **Error! Reference source not found.**).
j. If required, the specific issue of individual documents referenced (see 7.1).

7.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

7.4 Shock. Grade A shock qualification in accordance with MIL-S-901 is desired, but not required.

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7.5 Key words.

Charger, battery
Electrolyte, gelled
Lead-acid, regulated, valve
Transporter, pallet
Truck, lift
VRLA

MILITARY INTERESTS

Custodians:
Army – CR
Navy – SH
Air Force – 99

Review Activities:
Navy – AS, BC, EC, NW
Air Force – 03, 19, 71
DLA – GS, GS2

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FSS

Preparing Activity:
Navy – SH
(Project 6130-2008-001)

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